One acquisition/controller group with four independent dual-coil drivers | DRAFT

Logical eight-coil group, not a selected PCB module. Hardware watchdog gates each independent enable; current limiting remains open.

Power rails, coil terminations, Hall/host transport and programming connectors remain external interfaces.

Controller

- V_LOGIC
- GND
- SWCLK
- SWDIO
- RUN
- PWM_0_P
- PWM_0_N
- PWM_1_P
- PWM_1_N
- PWM_2_P
- PWM_2_N
- PWM_3_P
- PWM_3_N
- PWM_4_P
- PWM_4_N
- PWM_5_P
- PWM_5_N
- PWM_6_P
- PWM_6_N
- PWM_7_P
- PWM_7_N
- ENABLE_REQUEST_0
- ENABLE_REQUEST_1
- ENABLE_REQUEST_2
- ENABLE_REQUEST_3
- ENABLE_REQUEST_4
- ENABLE_REQUEST_5
- ENABLE_REQUEST_6
- ENABLE_REQUEST_7
- ADC_SCLK
- ADC_CS_N
- ADC_SDI
- ADC_SDO
- ADC_RESET_N
- ADC_ALARM
- AWAKE
- FAULT_N
- HOST_RX
- HOST_CS_N
- HOST_SCLK
- HOST_TX
- WATCHDOG_PULSE
- SAFE_OK
- ARM_REQUEST
- HOST_SYNC
- HALL_RX
- HALL_CS_N
- HALL_SCLK
- HALL_TX
- TEMP_DRIVER
- TEMP_COIL
- V_MOTOR_MONITOR
- POWER_FAULT_N

File: controller.kicad_sch

Acquisition

- V_ANALOG
- V_LOGIC
- GND
- VREF_CURRENT
- ADC_SDI
- ADC_RESET_N
- ADC_ALARM
- ADC_SDO
- ADC_SCLK
- ADC_CS_N
- CURRENT_0
- CURRENT_1
- CURRENT_2
- CURRENT_3
- CURRENT_4
- CURRENT_5
- CURRENT_6
- CURRENT_7

File: acquisition.kicad_sch

Protection

- V_LOGIC
- GND
- WATCHDOG_PULSE
- ARM_REQUEST
- POWER_FAULT_N
- FAULT_N
- SAFE_OK
- ENABLE_REQUEST_0
- ENABLE_REQUEST_1
- ENABLE_REQUEST_2
- ENABLE_REQUEST_3
- ENABLE_REQUEST_4
- ENABLE_REQUEST_5
- ENABLE_REQUEST_6
- ENABLE_REQUEST_7
- ENABLE_0
- ENABLE_1
- ENABLE_2
- ENABLE_3
- ENABLE_4
- ENABLE_5
- ENABLE_6
- ENABLE_7

File: protection.kicad_sch

Driver_0

- V_MOTOR
- V_ANALOG
- V_LOGIC
- GND
- VREF_CURRENT
- AWAKE
- FAULT_N
- PWM_0_P
- PWM_0_N
- ENABLE_0
- CURRENT_0
- COIL_0_P
- COIL_0_N
- PWM_1_P
- PWM_1_N
- ENABLE_1
- CURRENT_1
- COIL_1_P
- COIL_1_N

File: channel.kicad_sch

Driver_1

- V_MOTOR
- V_ANALOG
- V_LOGIC
- GND
- VREF_CURRENT
- AWAKE
- FAULT_N
- PWM_2_P
- PWM_2_N
- ENABLE_2
- CURRENT_2
- COIL_2_P
- COIL_2_N
- PWM_3_P
- PWM_3_N
- ENABLE_3
- CURRENT_3
- COIL_3_P
- COIL_3_N

File: channel.kicad_sch

Driver_2

- V_MOTOR
- V_ANALOG
- V_LOGIC
- GND
- VREF_CURRENT
- AWAKE
- FAULT_N
- PWM_4_P
- PWM_4_N
- ENABLE_4
- CURRENT_4
- COIL_4_P
- COIL_4_N
- PWM_5_P
- PWM_5_N
- ENABLE_5
- CURRENT_5
- COIL_5_P
- COIL_5_N

File: channel.kicad_sch

Driver_3

- V_MOTOR
- V_ANALOG
- V_LOGIC
- GND
- VREF_CURRENT
- AWAKE
- FAULT_N
- PWM_6_P
- PWM_6_N
- ENABLE_6
- CURRENT_6
- COIL_6_P
- COIL_6_N
- PWM_7_P
- PWM_7_N
- ENABLE_7
- CURRENT_7
- COIL_7_P
- COIL_7_N

File: channel.kicad_sch

Sheet: /
File: control_group.kicad_sch

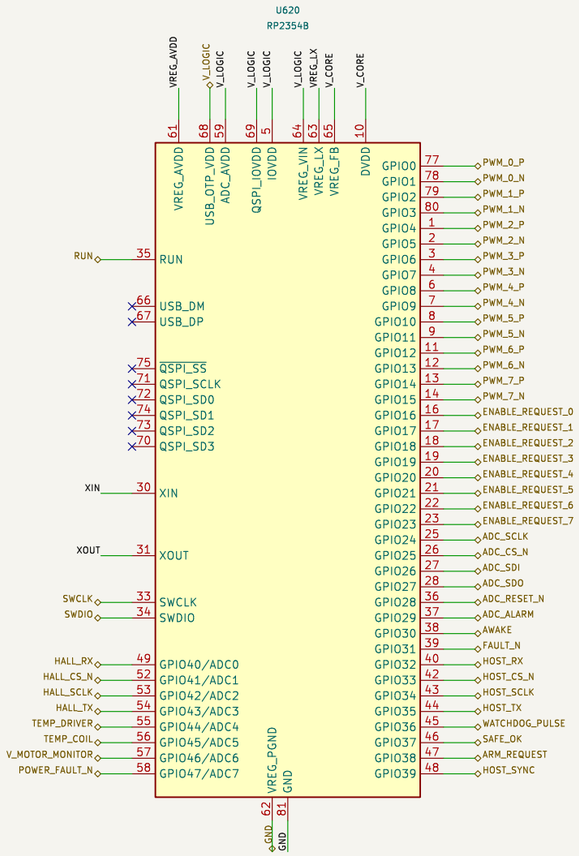
Title: Eight-coil electrical control group — DRAFT

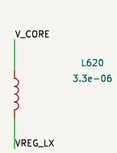
Size: A2	Date:	**Rev: working**
KiCad E.D.A. 9.0.8		Id: 8/8

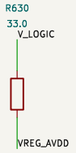
Local RP2354B controller | DRAFT | no manufacturing release

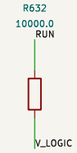
Eight paired PWM outputs; ADC acquisition uses PIO. ENABLE_REQUEST must pass through independent shutdown logic.

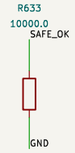
L20 pin 1 is the polarity-dot side and connects to V_CORE. Preserve manufacturer regulator placement and routing.



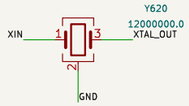








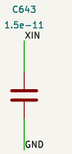






























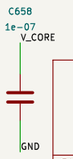












SWCLK / SWDIO / RUN: programming interface. USB and external QSPI pins intentionally unused; flash is in-package.

Sheet: /Controller/
File: controller.kicad_sch

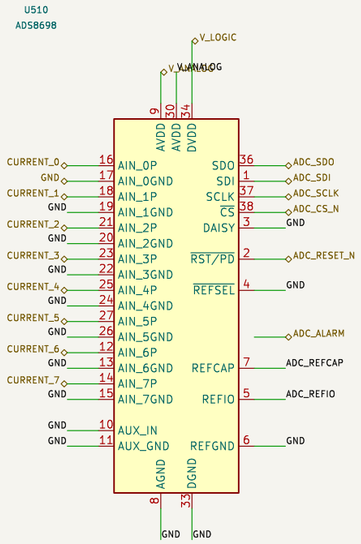
Title: Eight-coil local controller — DRAFT

Size: A3	Date:	Rev: working
KiCad E.D.A. 9.0.8		Id: 2/8

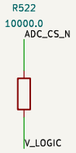
Shared eight-coil acquisition and bias reference | DRAFT | no manufacturing release

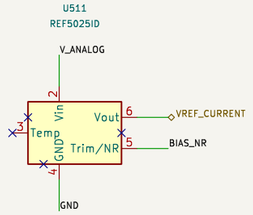
CURRENT inputs come from four dual-coil sheets. Supplies and synchronous SPI control remain external interfaces.

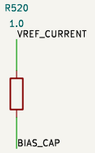
ADC internal reference serves only the ADC. REF5025 supplies the separate current-zero reference.





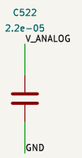






























Bulk capacitor effective value must remain above the datasheet minimum under DC bias and tolerance.

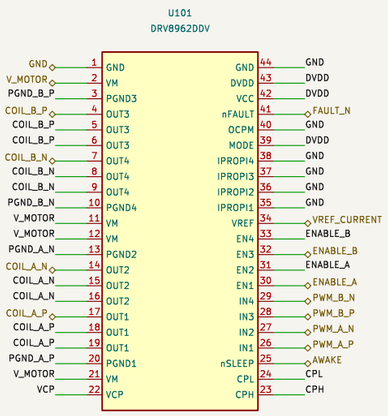
C20/C21 and C22/C23 belong at the two AVDD pin groups; keep all reference and ground return paths short.

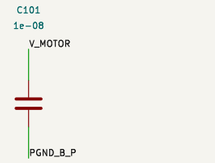
Sheet: /Acquisition/		
File: acquisition.kicad_sch		
Title: Shared current acquisition — DRAFT		
Size: A3	Date:	**Rev: working**
KiCad E.D.A. 9.0.8		Id: 3/8

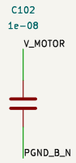
Dual-coil low-side reference | DRAFT | no manufacturing release

VREF uses the buffered current reference; V_LOGIC supplies the fault pullup. Shared interfaces remain open.

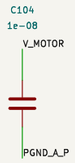
Each coil has two separate PGND shunts; the amplifier subtracts their voltages. Local return paths match the diagnostic topology.

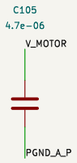












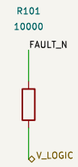




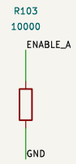


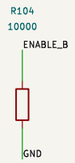


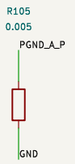


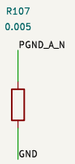


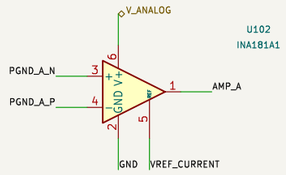


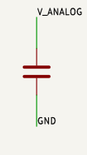


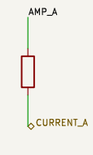










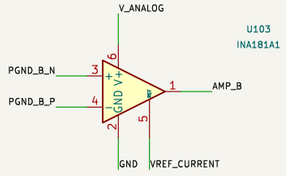




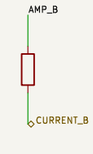
















Sheet: /Driver_0/
File: channel.kicad_sch

Title: Dual-coil low-side reference

Size: A3	Date:	**Rev: draft**
KiCad E.D.A. 9.0.8		Id: 4/8

Dual-coil low-side reference | DRAFT | no manufacturing release

VREF uses the buffered current reference; V_LOGIC supplies the fault pullup. Shared interfaces remain open.

Each coil has two separate PGND shunts; the amplifier subtracts their voltages. Local return paths match the diagnostic topology.

U201
DRV8962DDV

GND	1	GND	GND	44	GND
V_MOTOR	2	VM	DVDD	43	DVDD
PGND_B_P	3	PGND3	VCC	42	DVDD
COIL_B_P	4	OUT3	nFAULT	41	FAULT_N
COIL_B_P	5	OUT3	OCPM	40	GND
COIL_B_P	6	OUT3	MODE	39	DVDD
COIL_B_N	7	OUT4	IPROPI4	38	GND
COIL_B_N	8	OUT4	IPROPI3	37	GND
COIL_B_N	9	OUT4	IPROPI2	36	GND
PGND_B_N	10	PGND4	IPROPI1	35	GND
V_MOTOR	11	VM	VREF	34	VREF_CURRENT
V_MOTOR	12	VM	EN4	33	ENABLE_B
PGND_A_N	13	PGND2	EN3	32	ENABLE_B
COIL_A_N	14	OUT2	EN2	31	ENABLE_A
COIL_A_N	15	OUT2	EN1	30	ENABLE_A
COIL_A_N	16	OUT2	IN4	29	PWM_B_N
COIL_A_P	17	OUT1	IN3	28	PWM_B_P
COIL_A_P	18	OUT1	IN2	27	PWM_A_N
COIL_A_P	19	OUT1	IN1	26	PWM_A_P
PGND_A_P	20	PGND1	nSLEEP	25	AWAKE
V_MOTOR	21	VM	CPL	24	CPL
VCP	22	VCP	CPH	23	CPH

C201
1e-08
V_MOTOR
PGND_B_P

C202
1e-08
V_MOTOR
PGND_B_N

C203
1e-08
V_MOTOR
PGND_A_N

C204
1e-08
V_MOTOR
PGND_A_P

C205
4.7e-06
V_MOTOR
PGND_A_P

C206
1e-06
VCP
V_MOTOR

C207
1e-07
CPH
CPL

C208
1e-06
DVDD
GND

C209
1e-07
DVDD
GND

C216
4.7e-06
V_MOTOR
PGND_B_P

R201
10000
FAULT_N
V_LOGIC

R202
10000
AWAKE
GND

R203
10000
ENABLE_A
GND

R204
10000
ENABLE_B
GND

R205
0.005
PGND_A_P
GND

R207
0.005
PGND_A_N
GND

U202
INA181A1
V_ANALOG
PGND_A_N
PGND_A_P
AMP_A
GND
VREF_CURRENT

C210
1e-07
V_ANALOG
GND

R209
200.0
AMP_A
CURRENT_A

C212
1e-07
CURRENT_A
GND

C213
1e-07
CURRENT_A
GND

R206
0.005
PGND_B_P
GND

R208
0.005
PGND_B_N
GND

U203
INA181A1
V_ANALOG
PGND_B_N
PGND_B_P
AMP_B
GND
VREF_CURRENT

C211
1e-07
V_ANALOG
GND

R210
200.0
AMP_B
CURRENT_B

C214
1e-07
CURRENT_B
GND

C215
1e-07
CURRENT_B
GND

Sheet: /Driver_1/
File: channel.kicad_sch

Title: Dual-coil low-side reference

Size: A3	Date:	**Rev: draft**
KiCad E.D.A. 9.0.8		Id: 5/8

Dual-coil low-side reference | DRAFT | no manufacturing release

VREF uses the buffered current reference; V_LOGIC supplies the fault pullup. Shared interfaces remain open.

Each coil has two separate PGND shunts; the amplifier subtracts their voltages. Local return paths match the diagnostic topology.

U301
DRV8962DDV

GND	1	GND	GND
V_MOTOR	2	VM	DVDD
PGND_B_P	3	PGND3	VCC
COIL_B_P	4	OUT3	nFAULT → FAULT_N
COIL_B_P	5	OUT3	OCPM → GND
COIL_B_P	6	OUT3	MODE → DVDD
COIL_B_N	7	OUT4	IPROPI4 → GND
COIL_B_N	8	OUT4	IPROPI3 → GND
COIL_B_N	9	OUT4	IPROPI2 → GND
PGND_B_N	10	PGND4	IPROPI1 → GND
V_MOTOR	11	VM	VREF → VREF_CURRENT
V_MOTOR	12	VM	EN4 → ENABLE_B
PGND_A_N	13	PGND2	EN3 → ENABLE_B
COIL_A_N	14	OUT2	EN2 → ENABLE_A
COIL_A_N	15	OUT2	EN1 → ENABLE_A
COIL_A_N	16	OUT2	IN4 → PWM_B_N
COIL_A_P	17	OUT1	IN3 → PWM_B_P
COIL_A_P	18	OUT1	IN2 → PWM_A_N
COIL_A_P	19	OUT1	IN1 → PWM_A_P
PGND_A_P	20	PGND1	nSLEEP → AWAKE
V_MOTOR	21	VM	CPL → CPL
VCP	22	VCP	CPH → CPH

C301
1e-08
V_MOTOR
PGND_B_P

C302
1e-08
V_MOTOR
PGND_B_N

C303
1e-08
V_MOTOR
PGND_A_N

C304
1e-08
V_MOTOR
PGND_A_P

C305
4.7e-06
V_MOTOR
PGND_A_P

C306
1e-06
VCP
V_MOTOR

C307
1e-07
CPH
CPL

C308
1e-06
DVDD
GND

C309
1e-07
DVDD
GND

C316
4.7e-06
V_MOTOR
PGND_B_P

R301
10000
FAULT_N
V_LOGIC

R302
10000
AWAKE
GND

R303
10000
ENABLE_A
GND

R304
10000
ENABLE_B
GND

R305
0.005
PGND_A_P
GND

R307
0.005
PGND_A_N
GND

U302
INA181A1
V_ANALOG
PGND_A_N
PGND_A_P
AMP_A
GND
VREF_CURRENT

C310
1e-07
V_ANALOG
GND

R309
200.0
AMP_A
CURRENT_A

C312
1e-07
CURRENT_A
GND

C313
1e-07
CURRENT_A
GND

R306
0.005
PGND_B_P
GND

R308
0.005
PGND_B_N
GND

U303
INA181A1
V_ANALOG
PGND_B_N
PGND_B_P
AMP_B
GND
VREF_CURRENT

C311
1e-07
V_ANALOG
GND

R310
200.0
AMP_B
CURRENT_B

C314
1e-07
CURRENT_B
GND

C315
1e-07
CURRENT_B
GND

Sheet: /Driver_2/
File: channel.kicad_sch

Title: Dual-coil low-side reference

Size: A3	Date:	**Rev: draft**
KiCad E.D.A. 9.0.8		Id: 6/8

Dual-coil low-side reference | DRAFT | no manufacturing release

VREF uses the buffered current reference; V_LOGIC supplies the fault pullup. Shared interfaces remain open.

Each coil has two separate PGND shunts; the amplifier subtracts their voltages. Local return paths match the diagnostic topology.

U401
DRV8962DDV

GND	1	GND	GND	44	GND
V_MOTOR	2	VM	DVDD	43	DVDD
PGND_B_P	3	PGND3	VCC	42	DVDD
COIL_B_P	4	OUT3	nFAULT	41	FAULT_N
COIL_B_P	5	OUT3	OCPM	40	GND
COIL_B_P	6	OUT3	MODE	39	DVDD
COIL_B_N	7	OUT4	IPROPI4	38	GND
COIL_B_N	8	OUT4	IPROPI3	37	GND
COIL_B_N	9	OUT4	IPROPI2	36	GND
PGND_B_N	10	PGND4	IPROPI1	35	GND
V_MOTOR	11	VM	VREF	34	VREF_CURRENT
V_MOTOR	12	VM	EN4	33	ENABLE_B
PGND_A_N	13	PGND2	EN3	32	ENABLE_B
COIL_A_N	14	OUT2	EN2	31	ENABLE_A
COIL_A_N	15	OUT2	EN1	30	ENABLE_A
COIL_A_N	16	OUT2	IN4	29	PWM_B_N
COIL_A_P	17	OUT1	IN3	28	PWM_B_P
COIL_A_P	18	OUT1	IN2	27	PWM_A_N
COIL_A_P	19	OUT1	IN1	26	PWM_A_P
PGND_A_P	20	PGND1	nSLEEP	25	AWAKE
V_MOTOR	21	VM	CPL	24	CPL
VCP	22	VCP	CPH	23	CPH

C401
1e-08
V_MOTOR
PGND_B_P

C402
1e-08
V_MOTOR
PGND_B_N

C403
1e-08
V_MOTOR
PGND_A_N

C404
1e-08
V_MOTOR
PGND_A_P

C405
4.7e-06
V_MOTOR
PGND_A_P

C406
1e-06
VCP
V_MOTOR

C407
1e-07
CPH
CPL

C408
1e-06
DVDD
GND

C409
1e-07
DVDD
GND

C416
4.7e-06
V_MOTOR
PGND_B_P

R401
10000
FAULT_N
V_LOGIC

R402
10000
AWAKE
GND

R403
10000
ENABLE_A
GND

R404
10000
ENABLE_B
GND

R405
0.005
PGND_A_P
GND

R407
0.005
PGND_A_N
GND

U402
INA181A1
V_ANALOG
PGND_A_N
PGND_A_P
AMP_A
GND
VREF_CURRENT

C410
1e-07
V_ANALOG
GND

R409
200.0
AMP_A
CURRENT_A

C412
1e-07
CURRENT_A
GND

C413
1e-07
CURRENT_A
GND

R406
0.005
PGND_B_P
GND

R408
0.005
PGND_B_N
GND

U403
INA181A1
V_ANALOG
PGND_B_N
PGND_B_P
AMP_B
GND
VREF_CURRENT

C411
1e-07
V_ANALOG
GND

R410
200.0
AMP_B
CURRENT_B

C414
1e-07
CURRENT_B
GND

C415
1e-07
CURRENT_B
GND

Sheet: /Driver_3/
File: channel.kicad_sch

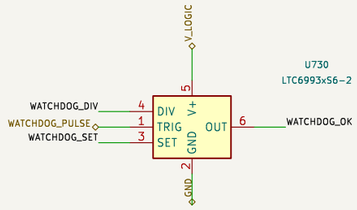
Title: Dual-coil low-side reference

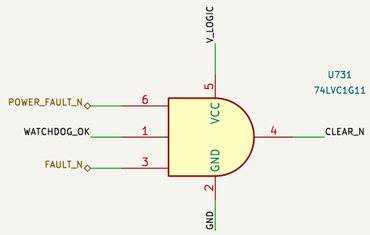
Size: A3	Date:	**Rev: draft**
KiCad E.D.A. 9.0.8		Id: 7/8

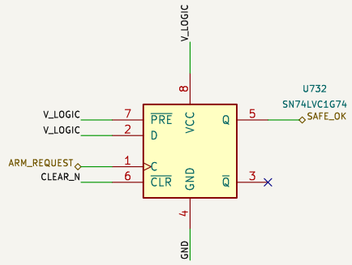
Independent enable protection | DRAFT | all rails and fault detectors must be completed

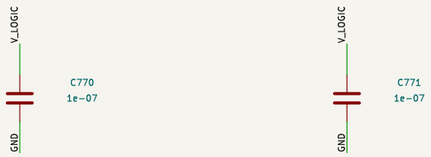
Watchdog pulses must follow validated sequential current updates. GPIO38 ARM_REQUEST requires explicit re-arm after any fault.

Timer and logic bounds are modeled; DRV8962 EN delay uses its specified maximum. No arbitrary overcurrent protection is claimed.









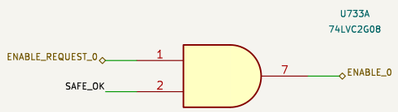


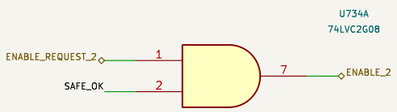


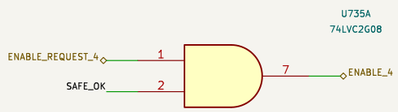


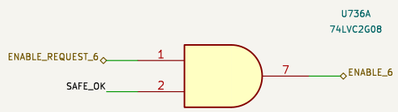


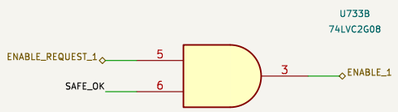


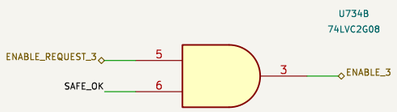


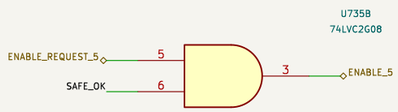


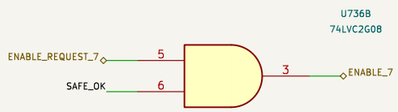


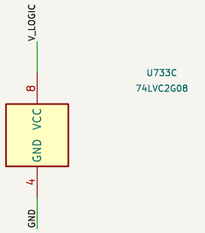


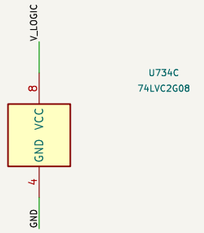


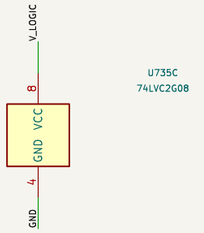


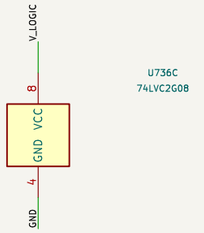




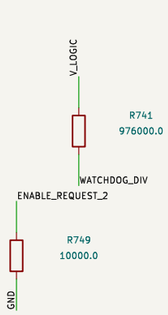


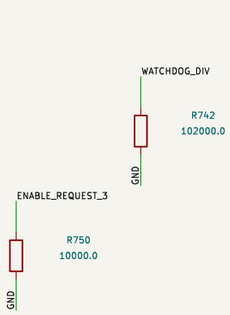






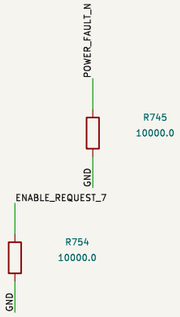


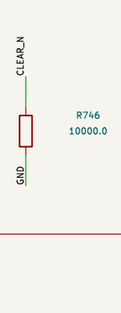


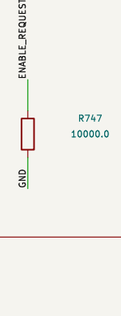


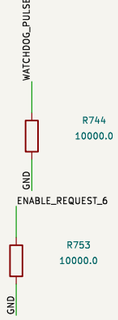












Sheet: /Protection/
File: protection.kicad_sch

Title: Independent current-control enable protection — DRAFT

Size: A3	Date:	**Rev: working**
KiCad E.D.A. 9.0.8		Id: 8/8